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Geometric origin of hierarchical patterns through history-dependent growth and arrest

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Abstract

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Keywords

Hierarchical pattern formation, Geometric exclusion, History-dependent growth, Growth arrest, Non-equilibrium pattern formation, Diffusion-free patterning

Geometric origin of hierarchical patterns through history-dependent growth and arrest

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Abstract

Hierarchical patterns are widespread in natural and synthetic systems, yet their formation is most often interpreted through diffusion-driven instabilities or static geometric constructions. While successful in many contexts, these frameworks struggle to encode growth history, temporal asymmetry, and pattern formation in non-diffusive, out-of-equilibrium systems. Here, I introduce the TRACE model (Temporal Recording through Arrested Coupled Evolution), a minimal framework in which pattern formation emerges from the competition between front propagation and kinetic arrest, without invoking diffusive transport. Growing domains leave geometric footprints that record their individual growth histories through exclusion. Numerical simulations reveal a broad spectrum of emergent patterns—including cell-like domains, stripes, banded structures, and hierarchical boundaries—arising from variations in spatial confinement, nucleation timing, and domain density. Crucially, the number and scale of patterns depend on accumulated domain interactions rather than nucleation alone. Finally, the model qualitatively reproduces key features observed in a minimal two-component chemical system exhibiting hierarchical macroscopic pattern formation, establishing a diffusion-free route to hierarchy encoded as a geometric record of growth history.

1. Introduction

Patterns such as periodic forms, colors, and rhythms are ubiquitous in nature across a wide range of length and time scales. These natural patterns have long inspired human efforts to address technological challenges, informing the design of buildings, materials, textiles,

transportation systems, and mechanical structures, as well as strategies to enhance material functionality and performance [1-5]. Understanding how such patterns emerge and evolve is therefore a central problem across many areas of science, as it provides fundamental insight into the mechanisms that govern structure formation in natural and engineered systems.

Alan Turing proposed the theoretical reaction–diffusion (RD) model to explain how spatially periodic patterns can emerge autonomously in chemical systems [6]. This concept, often referred to as pattern-forming instability, provides a powerful framework for understanding complex biological patterns, such as zebra stripes or lung branching. Thus, reaction–diffusion systems have been widely adopted to account for key aspects of morphogenesis, in which organized and heterogeneous structures arise through spontaneous symmetry breaking from an initially homogeneous state [7-11]. With the discovery of the CIMA reaction in the 1990s, which provided the first successful experimental realization of a reaction–diffusion system, the theoretical framework was validated, and a wide range of nonlinear behaviors, including chemical oscillations and spatiotemporal wave patterns, were revealed [8, 12-16]. While traditional Turing reaction-diffusion models have long been used to characterize patterns in spatially homogeneous systems, contemporary research is increasingly focused on explaining the emergence of spatial heterogeneity and complex structures through mechanisms such as mode mixing, which accounts for the simultaneous interaction of multiple Turing modes [17]. To explain hierarchical architectures such as the sequential stacking and multiple length scales reminiscent of molluscan shells-recent research has integrated growth history with mechanical patterns to elucidate the formation of 3D layered structures beyond simple repetitive motifs. This approach reveals that complex architectures emerge from diffusion-controlled growth-front instabilities, where anisotropic growth dictates the evolution of structural hierarchy [18].

Voronoi tessellation provides a powerful geometric framework for representing complex natural patterns by reducing irregular structures to a set of spatially partitioned domains defined by nucleation points. This abstraction has been widely used to model stochastic biological and material architectures, where the spatial arrangement of seeds determines cell size, boundary geometry, and global heterogeneity. By controlling the distribution and density of nucleation sites, Voronoi-based models enable systematic exploration of spatial organization without prescribing detailed microscopic dynamics. Owing to its geometric simplicity and generality,

Voronoi tessellation has been widely employed to interpret a broad range of natural and engineered systems. In biological contexts, it has been used to describe cellular packing and tissue organization, including epithelial growth and bone-like architectures. In materials science, Voronoi-based representations have provided insights into the cellular morphology of foams, the grain structure of polycrystalline metals, and the statistical geometry of disordered alloys. In these systems, spatial organization is often approximated through nucleation-driven partitioning, capturing essential features of heterogeneity at the mesoscale [19-21]. Despite their success in describing a wide range of natural patterns, both reaction–diffusion frameworks and Voronoi-based geometric models share fundamental limitations when confronted with hierarchical and history-dependent structures. In classical reaction–diffusion systems, pattern formation arises from diffusion-driven instabilities under continuous transport and supply of reacting species. While this mechanism successfully explains periodic motifs, it becomes increasingly difficult to account for hierarchical architectures spanning multiple length scales without introducing additional fields, nonlinear couplings, or finely tuned parameters. Voronoi tessellation, on the other hand, provides an efficient geometric description of spatial partitioning generated by nucleation events, and has been widely used to represent cellular and polycrystalline organizations. However, Voronoi constructions are inherently static: they encode only the final spatial arrangement of domains, while discarding the temporal sequence of growth, interaction, and arrest that led to the observed structure.

As a result, neither framework naturally captures (i) the emergence of hierarchical internal organization within domains, (ii) the explicit role of temporal growth history, or (iii) the irreversibility of pattern formation that prevents reconstruction of the underlying dynamics from the final geometry alone. While reaction–diffusion models rely on continuous diffusive transport and instability-driven wavelength selection, Voronoi-based descriptions are inherently geometric and static, prescribing spatial partitioning without an underlying temporal evolution. Consequently, both approaches lack a direct mechanism to encode growth history into space, and are ill-suited to describe pattern formation in non-diffusive, kinetically arrested, and strongly non-equilibrium systems governed by competing growth fronts rather than material transport. These limitations suggest the need for a minimal framework in which patterns are not prescribed by diffusive instabilities or static tessellations, but instead emerge as spatial records of growth histories shaped by geometric exclusion and kinetic arrest.

Experimental Phenomenology and Motivation

This study is motivated by a minimal two-component chemical system of phytic acid and methyltrimethoxysilane (MTMS) which exhibits spontaneous formation of complex hierarchical macroscopic patterns upon exposure to ambient air, mediated by a metastable intermediate state [22]. Initially prepared through a controlled mixing and heating process under vacuum, the precursor solution exists as a metastable, optically homogeneous phase that remains stable until atmospheric moisture initiates the reaction. Remarkably, without any externally imposed gradients, mechanical forcing, or pre-patterned substrates, the system reproducibly develops spatially organized surface morphologies as the sol-gel condensation proceeds under ambient conditions. As illustrated in the experimental observations (Figure 1), structural features typically nucleate from fixed positions at the droplet perimeter and propagate inward over time, resulting in morphologies consisting of interconnected labyrinthine or radially oriented domains. These patterns display a clear hierarchical organization across multiple length scales and, unlike many chemical oscillations, they are not transient; once formed, they become permanently recorded in the solidified structure.

Several key phenomenological features distinguish this system from classical reaction–diffusion (RD) experiments such as the CIMA or Belousov–Zhabotinsky reactions. Most notably, the pattern evolution is not sustained by continuous diffusion-driven feedback between chemical species. Instead, the ongoing sol–gel reaction leads to a rapid increase in viscosity as the siloxane network develops, which effectively suppresses long-range diffusion. Consequently, the evolving morphology becomes kinetically arrested, making the final pattern a static record of the growth history rather than a dynamic standing wave. Furthermore, quantitative analysis of the macroscopic structures reveals that the characteristic spacing of the surface wrinkles scales directly with the film thickness rather than with any intrinsic chemical diffusion length. This observation indicates that the dominant length scale is set geometrically, consistent with mechanical stress relaxation and buckling during solidification. In addition to this macroscopic organization, finer anisotropic textures are observed within individual domains, suggesting that local molecular ordering is imprinted during the phase separation process. These microscopic features remain spatially correlated with the larger-scale pattern, further reinforcing the notion that the observed hierarchy arises from the cumulative history of growth, chemical segregation, and mechanical stress.

Taken together, these experimental observations define a class of pattern formation phenomena

characterized by fixed nucleation positions, the suppression of diffusion during growth, and the irreversible recording of hierarchical spatial organization. These unique features provide the primary motivation for developing a minimal theoretical description that captures pattern formation as a purely geometric process, independent of complex diffusion kinetics or explicit mechanical constitutive laws.

Proposed Theoretical Framework and Modeling Strategy

The primary objective of this work is to establish a minimal theoretical framework that captures the essential features of the experimentally observed pattern formation within a two-dimensional setting. To achieve this, the model is intentionally constructed at a coarse-grained, macroscopic level, prioritizing the propagation of phase boundaries and competition for spatial occupancy over microscopic chemical or mechanical complexities. Consequently, several classes of physical processes are deliberately excluded to maintain the model's simplicity and focus. Specifically, the specific chemical identities of molecular species and detailed reaction pathways are not considered. While these factors influence the experimental system at a microscopic level, their primary macroscopic effect is to modulate the spatial distribution and density of nucleation sites, which this framework incorporates implicitly through the spatial and temporal statistics of domain nucleation without invoking explicit chemical kinetics. Similarly, mechanical stresses, elastic constitutive laws, and the formation of residual byproducts are not treated explicitly in this description. Although mechanical stress relaxation and buckling contribute to the detailed morphology observed in real materials, their dominant effect at the scale of interest is to irreversibly block further propagation once a domain has formed. This effect is represented phenomenologically through geometric exclusion and the arrest of growth, thereby avoiding the introduction of additional mechanical fields. Furthermore, crystalline anisotropy and orientation-dependent growth are neglected in favor of assuming isotropic propagation normal to the domain boundary. This simplification reflects the objective of isolating the fundamental roles of geometry and history accumulation rather than material-specific anisotropic effects.

Under these assumptions, the model does not attempt to describe the microscopic motion of individual particles or molecules but instead focuses on capturing the minimal ingredients required to reproduce the observed phenomenology. These core elements include fixed and asynchronous nucleation, irreversible front propagation, mutual exclusion between domains, and a strong dependence on growth history. In this sense, the framework represents a purely

geometric description of pattern formation that remains independent of diffusion, detailed chemistry, or explicit mechanical constitutive relations. It is important to note that, while the experimental system was previously discussed in the context of reaction-induced phase separation, the phenomenology addressed here can be understood and reproduced within a diffusion-free, geometric framework. This perspective motivates the formulation of the following minimal model as an experimentally tractable platform for understanding kinetically arrested patterning mechanisms.

Temporal Recording through Arrested Coupled Evolution (TRACE) Model

The experimental phenomenology described above require fundamentally different theoretical description from conventional diffusion-based or energetic-based pattern formation models. In this system, patterns do not emerge from a dynamic balance of competing fluxes, but rather from the irreversible occupation of space by domains that nucleate at fixed locations and grow over time. As a result, the final morphology act as a historical record of the order in which domains appear and expand, rather than a steady-state solution of underlying transport equations. The essential characteristic that should be captured is not the local dynamics of material transport, but the competition between domains for available space under irreversible growth. In particular, the hierarchical patterns observed in the experiment reflect a geometric time map. The domains that nucleated earlier establish extended territories, while the domains that appear later are constrained to fill the remaining interstitial regions. Once a region has undergone phase transformation, further growth into that region is permanently prohibited.

Motivated by these observations, I introduce the TRACE model (Temporal Recording through Arrested Coupled Evolution), a minimal framework that reveals the geometric origin of hierarchical pattern formation as a spatial record of arrested front evolution, without invoking diffusive transport. It represent (i) fixed-position nucleation, (ii) asynchronous domain generation, (iii) normal propagation of phase boundaries, and (iv) mutual exclusion between growing domains. The model aims to capture hierarchical pattern formation using a minimal set of fields and the complex spatiotemporal history of domain growth is reduced to a single effective variable that records geometric precedence in time. In the following, I formulate this model in two dimensions and show that it reproduces the essential features of the experimentally observed patterns without invoking diffusion, reaction kinetics, or explicit mechanical interactions.

To describe the geometric competition between growing domains, the system is represented by two coupled scalar fields defined over space and time. The first field, $\phi(x,t)$, is a level-set function whose zero contour defines the evolving boundary of transformed regions. The second field, $A(x,t)$, represents a local activity or growth-permission field that encodes whether further propagation is allowed at a given location. Domain growth occurs along the local normal to the interface, with a velocity proportional to $A(x,t)$. Specifically, the evolution of the level-set field is governed by an Eikonal-type equation: [23, 24]

$$\frac{\partial \phi}{\partial t} = -V_0 A |\nabla \phi|$$

Importantly, I abstain from the standard re-initialization of the level-set function. This allows the magnitude of ϕ to retain information about the elapsed time since the interface passed a given point. The activity field $A(x,t)$ evolves according to where α is a positive constant and H denotes the Heaviside function.

$$\frac{\partial A}{\partial t} = -\alpha A H(-\phi)$$

This term irreversibly suppresses activity in regions that have already undergone transformation, thereby enforcing geometric exclusion between competing domains. Together, these equations define a diffusion-free model in which hierarchical pattern formation emerges from asynchronous nucleation, irreversible front propagation, and mutual exclusion mediated purely by geometry and growth history. The scalar field $\phi(x,t)$ is introduced as a level-set function, with its zero contour defining the propagating boundary between transformed and untransformed regions. By construction, $\phi > 0$ corresponds to regions not yet transformed, while $\phi < 0$ denotes regions that have already undergone phase transformation. Unlike conventional level-set implementations, the field ϕ is not reinitialized during evolution. Consequently, its magnitude captures the time elapsed since the interface passed a given point, providing a geometric record of growth history. In this framework, more negative values of ϕ indicate earlier transformation, while values near zero represent more recently formed structures.

This interpretation frames the observed final patterns as a geometric time map. Domains that nucleate earlier propagate over greater distances to establish extended territories, whereas later

domains are geometrically confined by those already in place. Notably, this hierarchy emerges implicitly from the spatial structure of ϕ , without requiring diffusion, energetic competition, or explicit temporal rules. The field $A(x,t)$ is introduced to represent the irreversible exclusion associated with phase transformation. Physically, it encodes whether further propagation is permitted at a given location. Once a region has undergone transformation, the activity is locally suppressed and cannot be restored, reflecting the permanent blocking of growth observed in the experimental system. Through its coupling to the level-set field ϕ , the activity field ensures that domains do not invade regions that have already transformed. As a result, mutual exclusion between growing domains emerges naturally, without the need to impose additional geometric constraints or interaction rules. In this way, exclusion arises as a direct consequence of phase transformation, rather than as an independent mechanism.

Numerical realization of the TRACE model

I first validate the minimal model by considering the growth of a single domain nucleated at a fixed location. The purpose of this simulation is not to analyze numerical accuracy or algorithmic performance, but to verify that front propagation and irreversible arrest are consistently implemented within the proposed framework.

Figure 2 shows the temporal evolution of the level-set field ϕ and the activity field A for a single nucleation event. Following nucleation, the interface propagates isotropically, resulting in a circular growth front. As this front advances, regions that undergo transformation exhibit increasingly negative values of ϕ , providing a spatial record of the time elapsed since solidification. Simultaneously, the activity field A is locally suppressed behind the propagating front, creating a frozen core where further evolution is inhibited. This irreversible decay of activity ensures that once a region has transformed, it remains stable and inactive. Notably, no external constraints or complex interaction rules are required; this self-exclusion emerges naturally from the coupling between ϕ and A . This single-domain simulation validates the model's fundamental consistency and serves as a reference for the multi-domain and hierarchical patterns described in subsequent sections.

Figure 3(a) demonstrates that the proposed growth–arrest mechanism remains consistent in a multi-domain setting. Multiple domains initiated simultaneously expand via front propagation until neighboring fronts collide, where growth is locally arrested. This arrest-induced exclusion arises without any explicit interaction rules and results in a stable spatial partitioning. The

resulting pattern represents a geometric record of irreversible space occupation rather than diffusion-driven instability, confirming that collective patterns emerge from the identical minimal dynamics as the single-domain case. Notably, domain growth does not always proceed to full space-filling. If arrested at intermediate stages, isolated circular morphologies naturally emerge. This indicates that the final patterns reflect both geometric exclusion and the extent of propagation recorded prior to arrest. Figures 3(b) and (c) demonstrate the influence of domain density by varying the number of simultaneous nucleation sites while keeping all other parameters constant. In both the low-density (5×5) and high-density (15×15) arrays, the activity field A does not fully decay to zero, indicating the persistence of residual active regions. This behavior stems from the geometric competition between propagating fronts initiated at identical times. When the spacing between domains is incommensurate with the characteristic arrest length scale, the fronts fail to block each other synchronously, leaving behind active boundaries. As a result, grid-locked defects reminiscent of grain-boundary networks emerge, despite the absence of crystalline anisotropy or diffusive transport. Magnified views of ϕ reveal hierarchical structures emerge from the superposition of multiple domain histories, rather than from the interior of a single domain. This morphological sensitivity to domain density underscores that the final pattern is governed by the global geometric arrangement and the temporal history of nucleation, rather than by purely local dynamical processes.

To examine the role of solidification kinetics, the arrest rate α was varied while keeping all other parameters fixed (Figure 3(d)). As α decreases, the activity field A decays more slowly allowing the propagation dynamics to persist over extended duration. Consequently, boundary regions become increasingly dominant in the spatial morphology. This delayed kinetic arrest enables the solidification front to record a broader temporal history before being frozen, leading to the emergence of more intricate, hierarchical patterns. These results indicate that the final morphology is determined not only by the initial geometric arrangement of domains but also significantly by the internal kinetics of the solidification process. Figures 3(e–g) investigate the effect of asynchronous domain nucleation by introducing temporal and spatial disorder in the initiation of domains. Remarkably, even for a high domain density (15×15 array), asynchronous nucleation leads to complete arrest throughout the system, in contrast to the synchronized case where persistent active boundaries remain. In this regime, early-nucleated domains establish extended exclusion zones that geometrically constrain the growth of subsequent domains, effectively preventing grid-locking and facilitating more complete space filling. When both nucleation times and positions are randomized, the resulting morphology becomes strongly

asymmetric, reflecting the non-uniform superposition of domain histories. These results demonstrate that the final pattern is governed not only by geometric arrangement and arrest kinetics, but also critically by the temporal ordering of domain nucleation events.

Figure 4(a) demonstrates that spatially confined nucleation is sufficient to generate ordered stripe-like patterns within the proposed front-arrest model. Domain seeds are synchronously introduced along a narrow boundary region, while the governing dynamics remain isotropic omitting both diffusion and reaction terms. Following nucleation, stripe-like structures emerge perpendicular to the propagation direction. These patterns do not arise from bulk instabilities or wavelength selection, but from the temporal ordering of front arrival and space occupation. Each stripe effectively records a distinct stage of the advancing solidification front. This result confirms that ordered patterns can emerge solely from the interplay between front propagation and arrest-induced exclusion, without requiring diffusion, chemical kinetics, or imposed anisotropy. Figure 4(b) illustrates how a decreased arrest rate α fundamentally alters the boundary morphology. Rather than producing thin interfaces, a lower α allows the activity field A to persist for a longer duration after the initial front collision. This delayed kinetic arrest leads to a spatial broadening of the boundaries, where the interaction between neighboring fronts is recorded over a wider area. The resulting morphology exhibits strongly anisotropic, column-like structures. Notably, this anisotropy is not pre-defined by directional growth rules but emerges from the extended competition between propagating fronts coupled with geometric exclusion. By modulating the arrest rate, the model can effectively tune the width and spatial complexity of these arrested boundaries, transitioning from sharp interfaces to prominent hierarchical seams. Figure 4(c) reveals that the emergence of spatial patterns does not scale linearly with the number of nucleation sites. Instead, the characteristic boundary morphology and the resulting pattern scale are sensitively determined by the global domain spacing and the dynamic competition between neighboring fronts. This suggests that the final morphology is a product of global geometric constraints specifically the collective arrangement of fronts rather than a mere superposition of local nucleation events. As the seed density increases, the reduced spacing forces more frequent front collisions, effectively increasing the geometric resolution of the arrested structures. Figure 4(d) demonstrates that asynchronous nucleation can generate spatial organizations that resemble diffusion-mediated smoothing, despite the entire absence of diffusive transport or material exchange. This behavior arises from the prescribed temporal offsets in the growth histories of neighboring domains. When the initiation times are slightly staggered, the resulting superposition of fronts breaks the

sharp coherence seen in synchronized cases, leading to a geometric regularization of the pattern. These results confirm that diffusion-like appearances can emerge solely from the accumulation of purely geometric and temporal mechanisms, suggesting that the model can mimic complex physical morphologies through minimal dynamical rules.

Figure 4(e) examines the effect of spatial confinement on pattern formation by employing a high-aspect-ratio field geometry. Despite the absence of explicit anisotropic terms in the governing equations, the resulting morphology exhibits strongly directional, linear patterns. This demonstrates that the global boundary geometry serves as a primary determinant of the domain history. The constrained spatial extent biases front propagation and exclusion, forcing preferential growth orientations that emerge purely from geometric competition. Furthermore, when coupled with asynchronous nucleation, this confinement induces a transition to globally asymmetric patterns, even when initial nucleation positions are arranged symmetrically. These results underscore that directional organization in this model does not require additional interaction terms or directional couplings. Instead, anisotropy arises naturally from the deterministic interplay between spatial geometry, front propagation, and the accumulated history of exclusion. Figure 4(f) evaluates the model's ability to qualitatively replicate the key morphological features observed in the experimental system that motivated this study. Nucleation sites are distributed along the outer boundary of a circular domain, with no initial seeds in the interior. As the process evolves, the propagating fronts generate well-defined internal boundaries that advance centripetally toward the center. This result demonstrates that complex inward-growing patterns can emerge solely from geometric exclusion and history-dependent front propagation, without invoking diffusion, long-range interactions, or additional driving fields. The resulting morphology qualitatively resembles experimentally observed boundary-driven pattern formation, indicating that the proposed minimal model captures essential geometric aspects of the underlying process.

Discussion

The TRACE model defines pattern formation as a purely geometric process governed by the competition between front propagation and kinetic arrest, independent of diffusion, transport, or continuous material supply. All domains originate from discrete nucleation events at fixed

positions, with subsequent evolution proceeding through spatial exclusion generated by propagating phase fronts whose activity decays irreversibly. In this framework, delayed arrest does not introduce any effective transport mechanism, but instead prolongs the lifetime over which geometric exclusion operates. The hierarchical patterns emerging in this system do not reflect the internal ordering of individual domains. Instead, they represent a spatial record of multiple domain histories superimposed under non-uniform exclusion. The final morphology encodes when and where domains nucleated, how long their fronts remained active, and how geometric competition unfolded in space. This establishes the pattern hierarchy as a direct consequence of history-dependent exclusion rather than local dynamical instabilities.

Beyond validating its internal consistency, the TRACE model's numerical experiments explore how morphology responds to variations in spatial geometry, domain density, and nucleation timing, mirroring experimental parameter tuning. I find that the spatial extent and confinement act as intrinsic control parameters, fundamentally influencing boundary spacing and the persistence of active regions. In high-aspect-ratio or confined spaces, this geometric bias alone induces the emergence of linear, stripe-like, or banded structures without the need for explicit anisotropic terms. Temporal asynchrony in nucleation further modulates these effects. When initiation occurs at staggered intervals, the resulting "phase shifts" in growth histories allow surviving fronts to dominate the available space, leading to pronounced symmetry breaking even under symmetric initial positions. Consequently, the final patterns are determined not by nucleation density alone, but by the coupled history of space occupation and front survival. The model successfully reproduces a diverse range of morphologies including cellular domains, hierarchical subdivisions, and the centripetal, boundary-driven inward growth observed in our motivating experimental system. These results underscore that morphological diversity arises from the deterministic competition between geometry and time encoded in the field history, rather than from material-specific interactions.

These results show that a minimal description based solely on front propagation and kinetic arrest is sufficient to generate a broad class of nontrivial spatial patterns. Proposed model in this system demonstrates how geometry, nucleation history, and temporal competition alone can encode complex morphological outcomes without invoking diffusion, material transport, or internal domain ordering. The observed patterns are not the result of fine-tuned interactions, but emerge generically from the accumulation of exclusion events in space and time. This perspective suggests that arrested boundary formation may serve as a unifying mechanism for

understanding pattern diversity across systems where growth fronts compete under finite activity lifetimes. While this study focuses on a simplified numerical framework, the underlying principles are not tied to a specific material or length scale. The model thus provides a compact theoretical basis for interpreting pattern formation in systems where spatial confinement, asynchronous activation, and history-dependent exclusion play a dominant role.

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Data and Code Availability

The source code is available at: https://github.com/junhyunglee73/TRACE_Model.git

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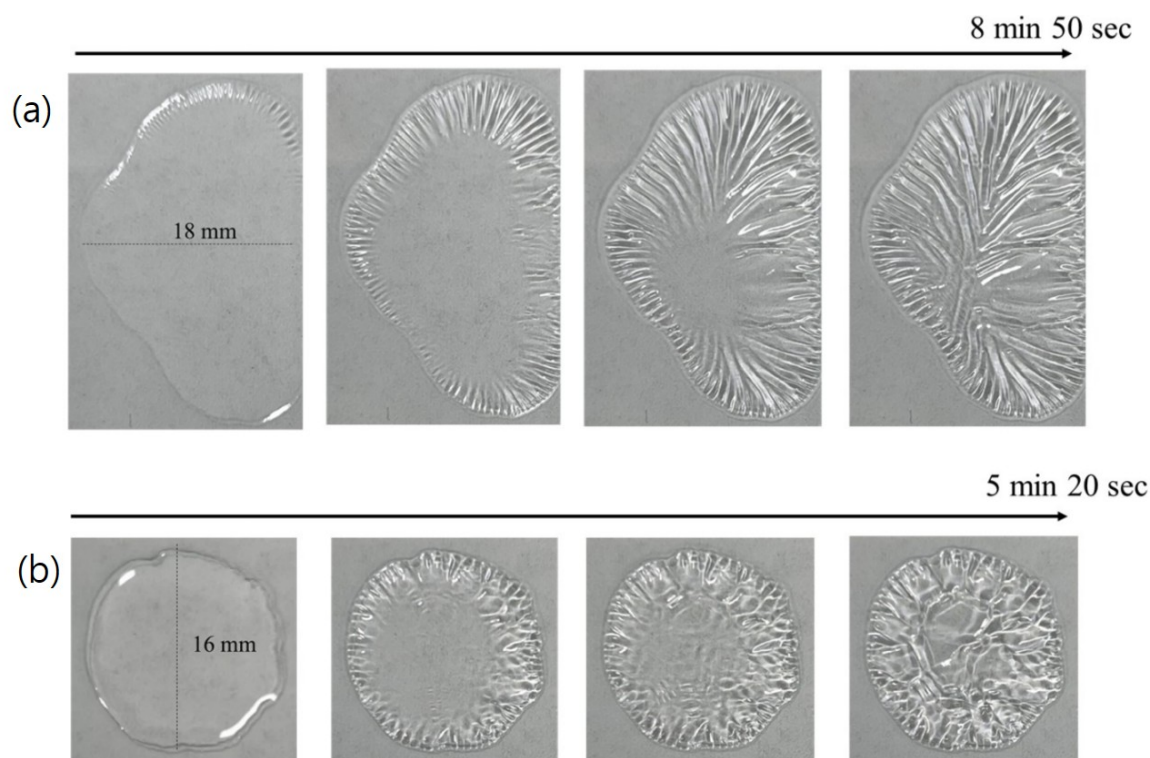


Figure 1. (a) Sequential optical images showing the gradual evolution of radial and labyrinthine features in an MSP-3 droplet on glass. (b) Corresponding time series for MSP-10, displaying finer and more interconnected patterns compared to MSP-3. *Adapted from Ref. [22].*

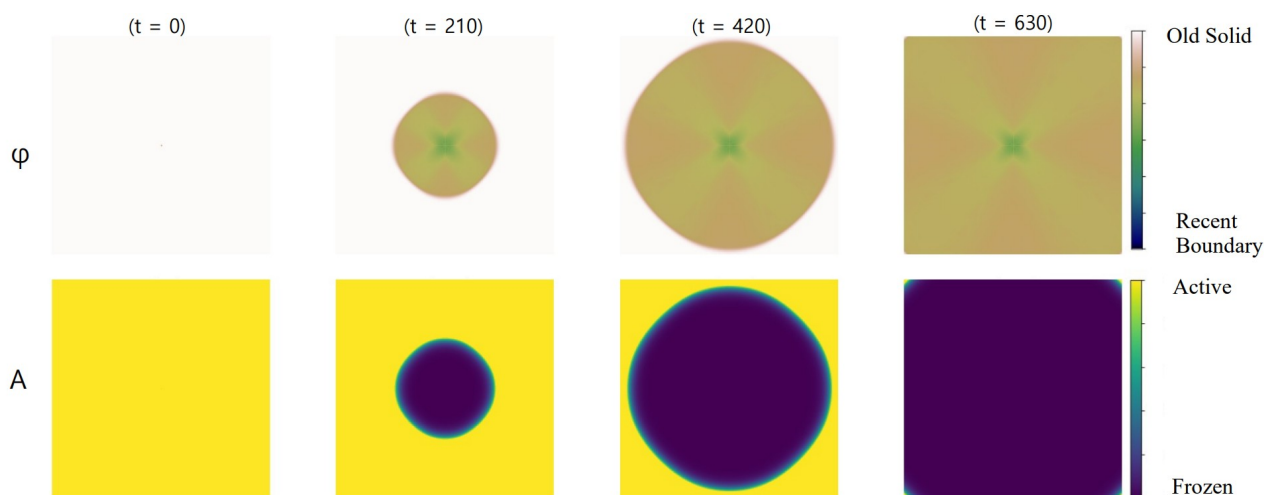


Figure 2. Numerical implementation of the minimal growth-arrest model initiated from a single nucleation site. Top row shows the evolution of the level-set field ϕ , representing the propagating growth front and its accumulated history. Bottom row shows the corresponding activity field A , illustrating the progressive arrest of growth behind the front. The simulation confirms that front propagation and kinetic arrest alone generate a well-defined exclusion region without diffusive transport.

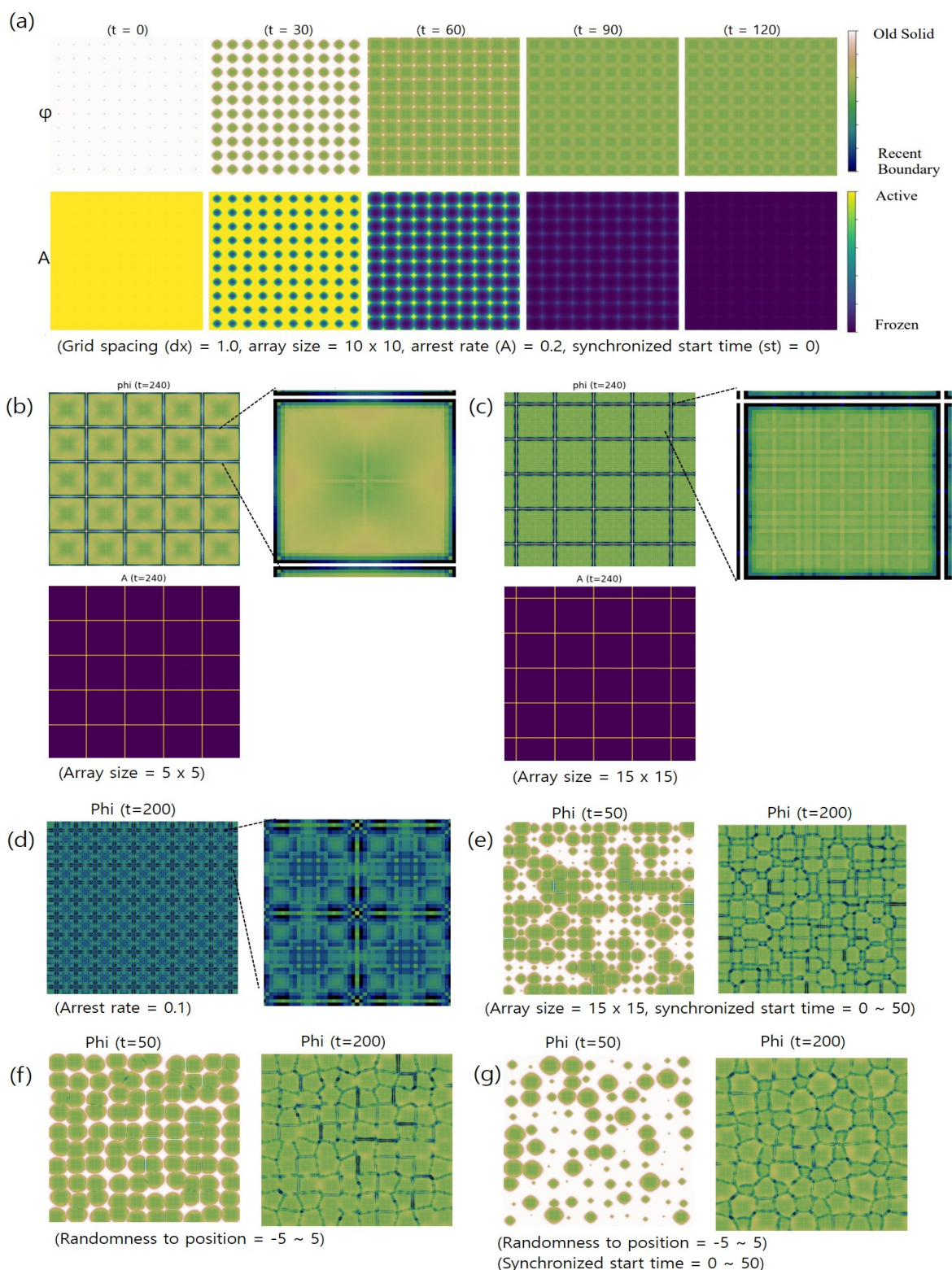


Figure 3. Pattern formation arising from multiple interacting domains. (a) Synchronously nucleated domains arranged on a regular lattice exhibit mutual geometric exclusion, leading to polygonal domain partitioning and arrested boundaries. (b–d) Variations in domain density and arrest rate modify the spatial extent of exclusion and the internal hierarchical structure of the patterns. (e–g) Asynchronous nucleation in time and space produces asymmetric and history-dependent morphologies, demonstrating that the final pattern reflects the temporal ordering of domain growth rather than local dynamics.

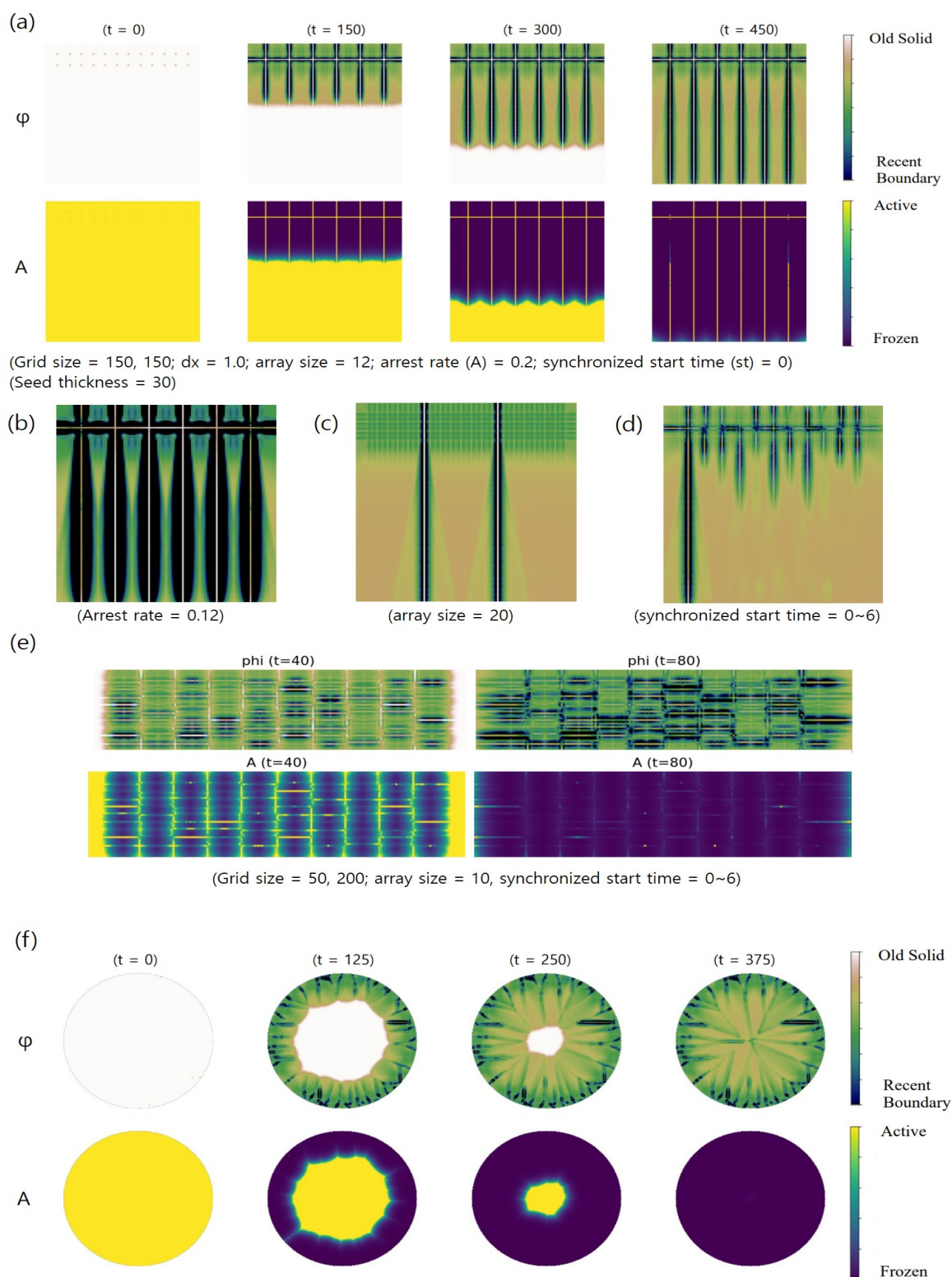


Figure 4. Emergence of diverse pattern morphologies under modified spatial and nucleation conditions. (a–d) Confinement of nucleation sites to a restricted region induces stripe-like and banded patterns through anisotropic geometric exclusion. (e) Strong spatial confinement enhances linear ordering and suppresses isotropic domain expansion. (f) Random nucleation along a circular boundary generates inward-propagating fronts and arrested internal structures, reproducing key phenomenological features observed in the experimental system.